LLMs & cognition

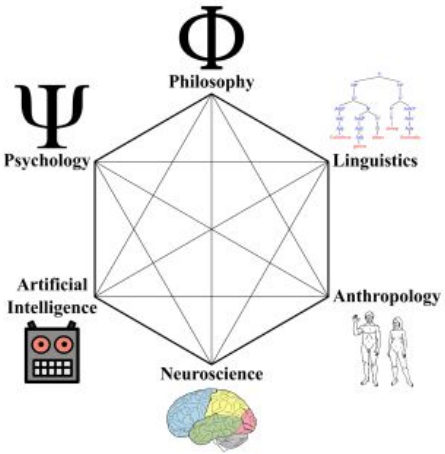
2. DATE DESPRE DISCIPLINĂ
COURSE IDENTIFICATION DETAILS

Motivație / Motivation:

From a theoretical standpoint, we are in the age of crossing to the other side of Turing's test. What will we find there? Should we fear the singularity? Can we improve AI tools and understand them better by taking inspiration from human cognition? Can models of human thought benefit from theories of computation?

This course aims to provide introductory notions in the mechanisms that drive Large Language Models (training, finetuning, attention, induction heads, explainability) and to connect them to different areas of Cognitive Science (Neuroscience, Psychology, Linguistics, Philosophy, and Anthropology). The course curriculum aims to cover practical applications in using and training LLMs in relation to electroencephalography, eye tracking, functional magnetic resonance imaging, and other physiological data.

The course will be a platform for discussions into questions that are now relevant to society as a whole and for learning to build tools around LLMs more responsibly.

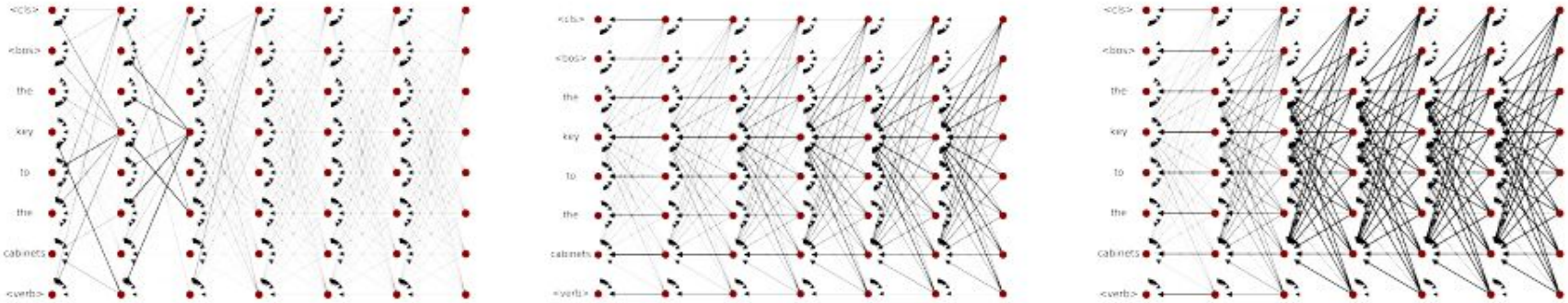


8.1. Curs Course	**Metode de predare** Teaching methods	**Observaţii** Remarks
Introduction to LLMs, Cognitive Science, Neuroscience, course structure, short history		
Vector spaces, semantic representations, N-gram language models, Tokenization	- prezentarea bazată pe slide-uri - explicația, descrierea și exemplificarea - slides on projector - explicative and descriptive talk	Resurse utilizate: - tablă de scris - proiector - computer Resources used: - whiteboard - projector - computer
Compression, Neural Language Models, RNNs, Transformers, State-Space Models		
Attention Flows, Explainability, Psycholinguistic features of LLMs		
EEG Data Analysis - Frequency Spectrum, Event Related Potentials, Artificial Neuron Activations		
Eye Tracking Data Analysis - Fixations, Saccades, Reading Times, Language Modelling Surprisal		
fMRI data analysis - Voxels, Human Language Network, Neuron Activations, LLMs for driving/suppressing the human language network		

Bibliografie/ Bibliography**:**

- *Build a Large Language Model (From Scratch)* Sebastian Raschka, MEAP August 2024, ISBN 9781633437166
- Neuromatch computational neuroscience course, https://compneuro.neuromatch.io/
- Eye-Tracking: A Guide for Applied Linguistics Research, Kathy Conklin, Ana Pellicer-Sánchez, Gareth Carrol; Cambridge University Press, 2018
- Mahowald, K., Ivanova, A. A., Blank, I. A., Kanwisher, N., Tenenbaum, J. B., & Fedorenko, E. (2024). *Dissociating language and thought in large language models*. In Trends in Cognitive Sciences. Elsevier BV. https://www.cell.com/action/showPdf?pii=S1364-6613%2824%2900027-5
- Kandel, E. R., Schwartz, J. H., Jessell, T. M., Siegelbaum, S. A., & Hudspeth, A. J. (2000). *Principles of neural science, fifth edition* (5th ed.). New York, NY: McGraw-Hill Medical.
- Tuckute, G., Kanwisher, N., & Fedorenko, E. (2024). Language in brains, minds, and machines. Annual Review of Neuroscience, 47.
- Zou, A., Phan, L., Chen, S., Campbell, J., Guo, P., Ren, R., ... & Hendrycks, D. (2023). Representation engineering: A top-down approach to ai transparency.
- Lelia A. Erscoi, Annelies Kleinherenbrink, and Olivia Guest; Pygmalion Displacement: When Humanising AI Dehumanises Women, https://osf.io/preprints/socarxiv/jqxb6
- Beinborn, Lisa, and Nora Hollenstein. *Cognitive Plausibility in Natural Language Processing*. Springer Nature, 2023.
- Proceedings of recent conferences and journals on AI and AI-adjacent topics (ACL, NeurIPS, ICLR, etc)

Interpreting LLMs: attention circuits



(a) Embedding attentions (b) Attention rollout (c) Attention flow

Figure 1: Visualisation of attention weights.

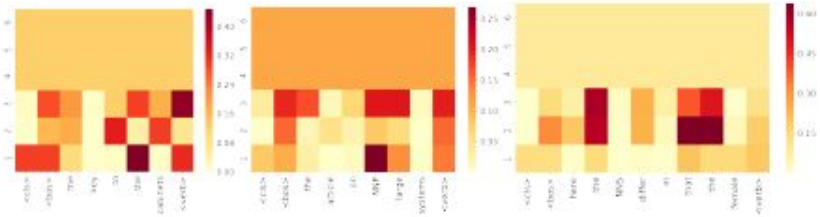
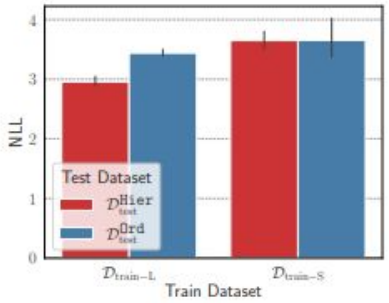


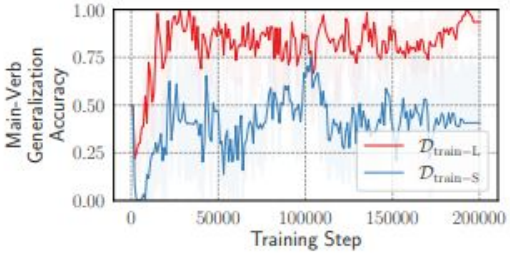
Figure 2: Raw Attention maps for the CLS token at

Abnar, S., & Zuidema, W. (2020, July). Quantifying Attention Flow in Transformers. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics* (pp. 4190-4197).

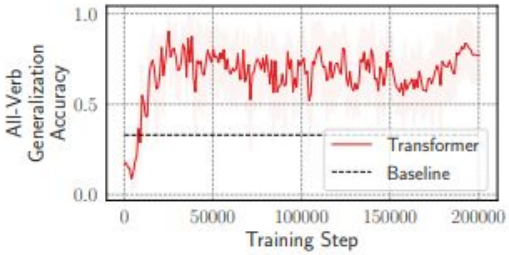
Syntax representation and generalization in LLMs



(a) Negative log-likelihood evaluated on the hierarchical ($\mathcal{D}_{\text{test}}^{\text{Hier}}$) and linear ($\mathcal{D}_{\text{test}}^{\text{Lin}}$) generalization test sets.

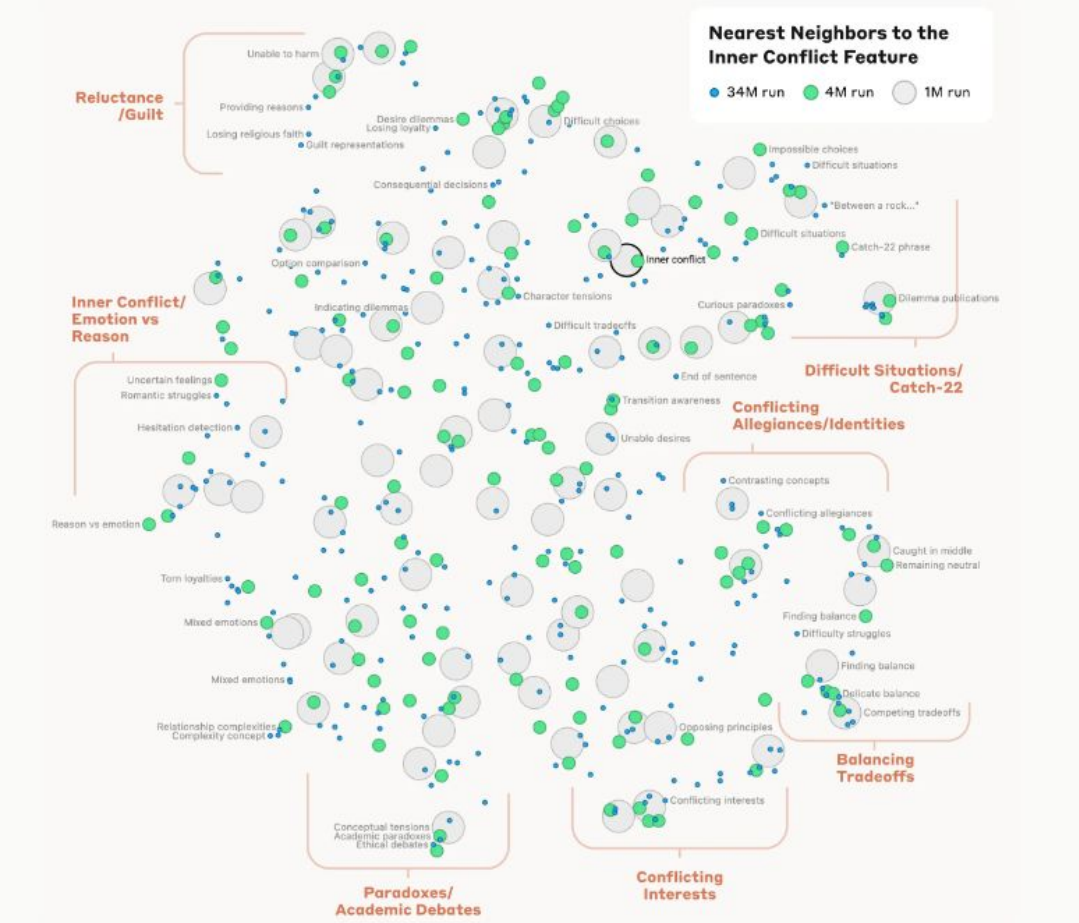


(b) Main-verb generalization accuracy on the $\mathcal{D}_{\text{test}}^{\text{Hier}}$ test set.



(c) All-verb generalization accuracy on $\mathcal{D}_{\text{test}}^{\text{Hier}}$ test for the model trained on the 120-types dataset.

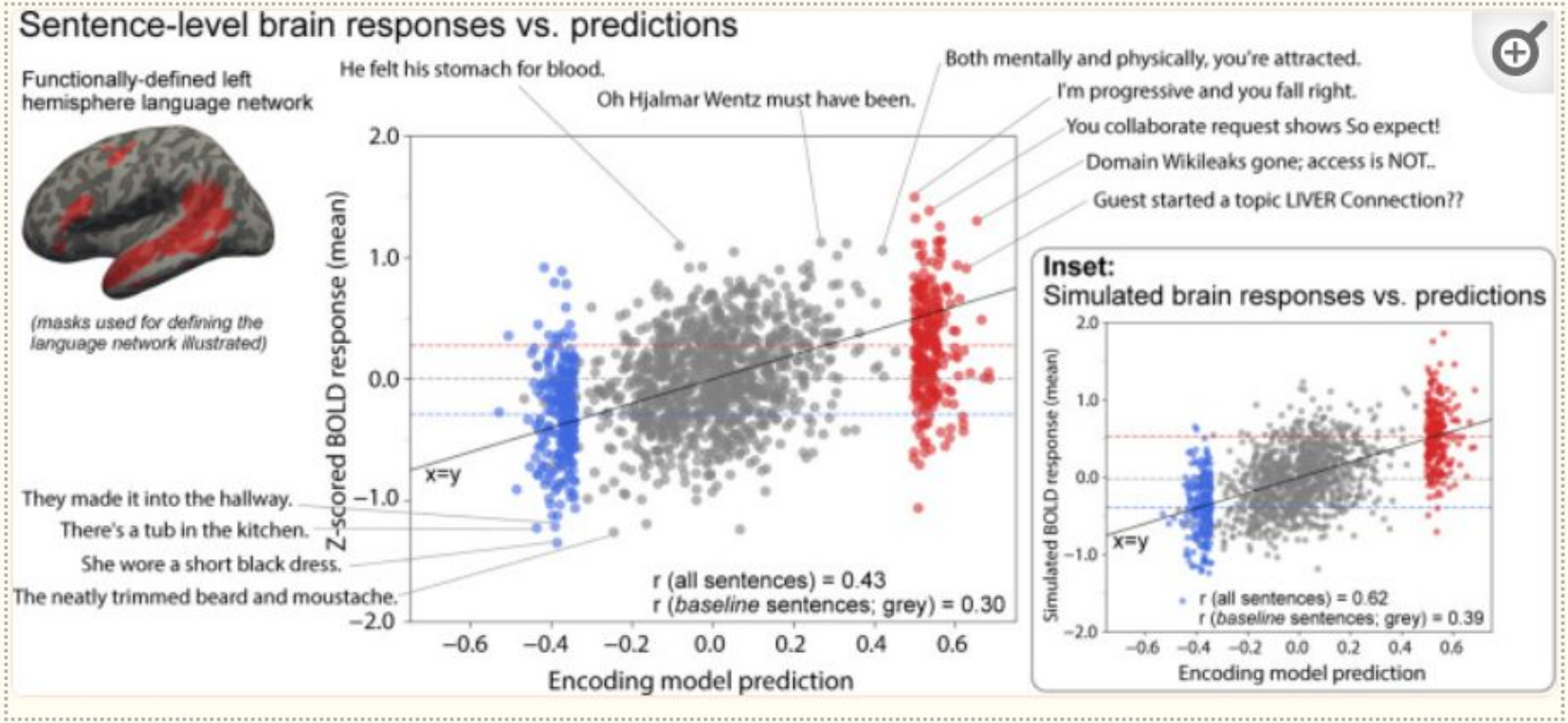
Ahuja, Kabir, et al. "Learning Syntax Without Planting Trees: Understanding When and Why Transformers Generalize Hierarchically." *arXiv preprint arXiv:2404.16367* (2024).



Mapping the mind of an LLM

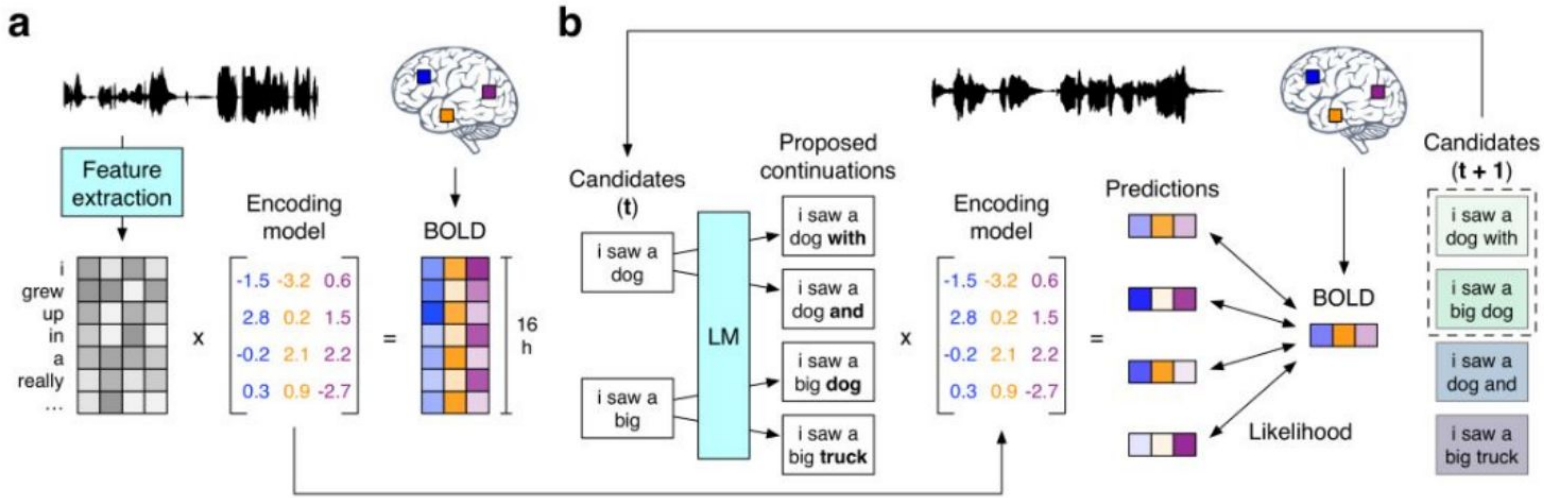
Templeton, Adly. *Scaling monosemanticity: Extracting interpretable features from claude 3 sonnet*. Anthropic, 2024.

fMRI BOLD response prediction via LLM



Tuckute, G., Sathe, A., Srikant, S., Taliaferro, M., Wang, M., Schrimpf, M., ... & Fedorenko, E. (2024). Driving and suppressing the human language network using large language models. *Nature Human Behaviour*, *8*(3), 544-561.

Reconstruction of language from fMRI



Tang, Jerry, et al. "Semantic reconstruction of continuous language from non-invasive brain recordings." *Nature Neuroscience* 26.5 (2023): 858-866.

LLM attention vs human attention

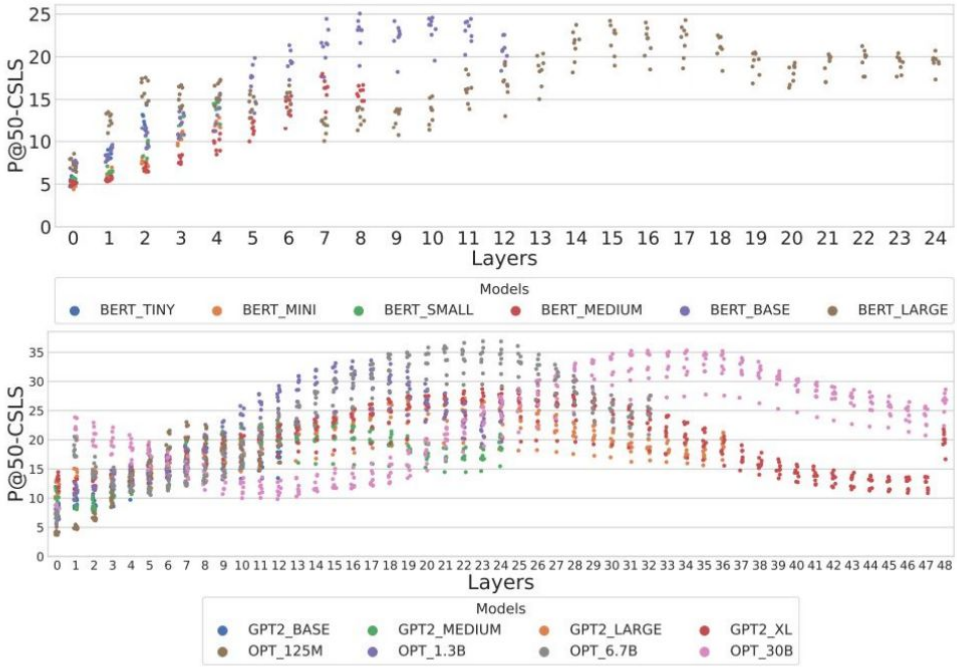
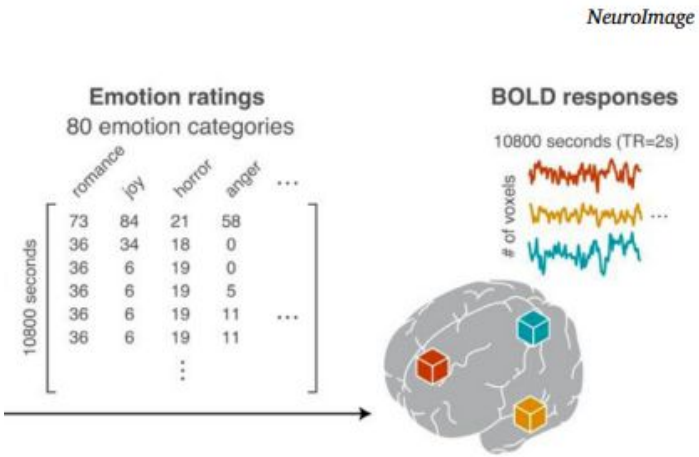
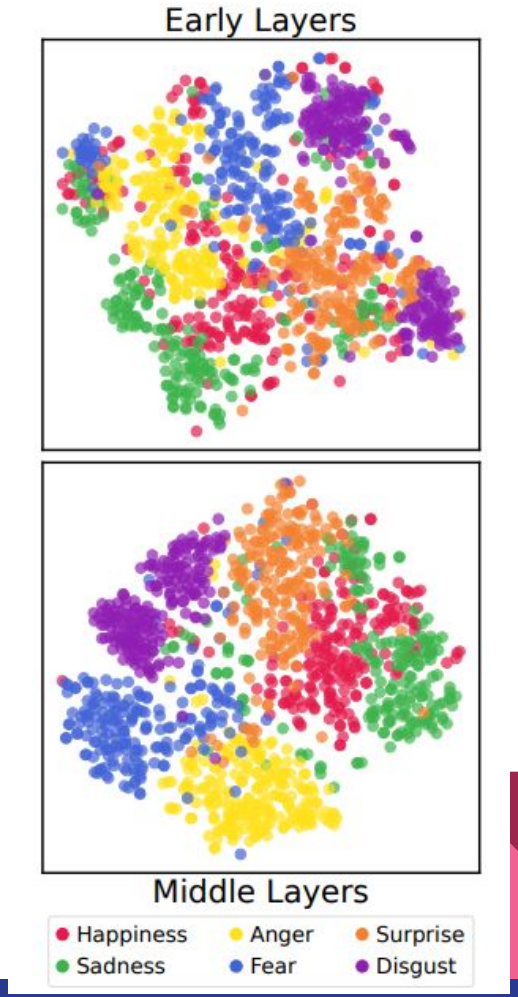


Figure 8: **Alignment precision results across layers.** The plot shows alignment with fMRI (Harry Potter dataset) improves with model depth for LLMs and Procrustes Analysis with Gaussian Random Projection.

Li, Jiaang, et al. "Structural similarities between language models and neural response measurements." *arXiv preprint arXiv:2306.01930* (2023).

Emotions in the brain and in LLMs

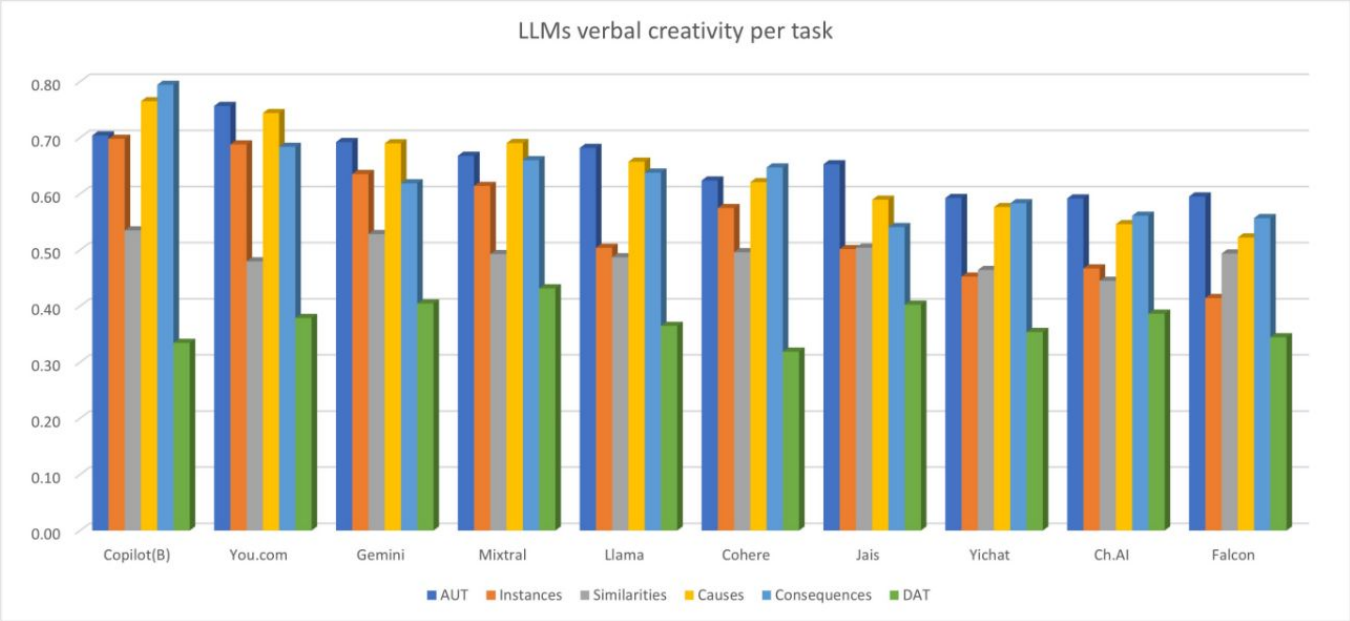




Koide-Majima, N., Nakai, T., & Nishimoto, S. (2020). Distinct dimensions of emotion in the human brain and their representation on the cortical surface. *NeuroImage*, *222*, 117258.

Zou, A., Phan, L., Chen, S., Campbell, J., Guo, P., Ren, R., ... & Hendrycks, D. (2023). Representation engineering: A top-down approach to ai transparency.

LLMs creativity



An integrated benchmark for verbal creativity testing of LLMs and humans, KES, 2024

Psychology of LLMs

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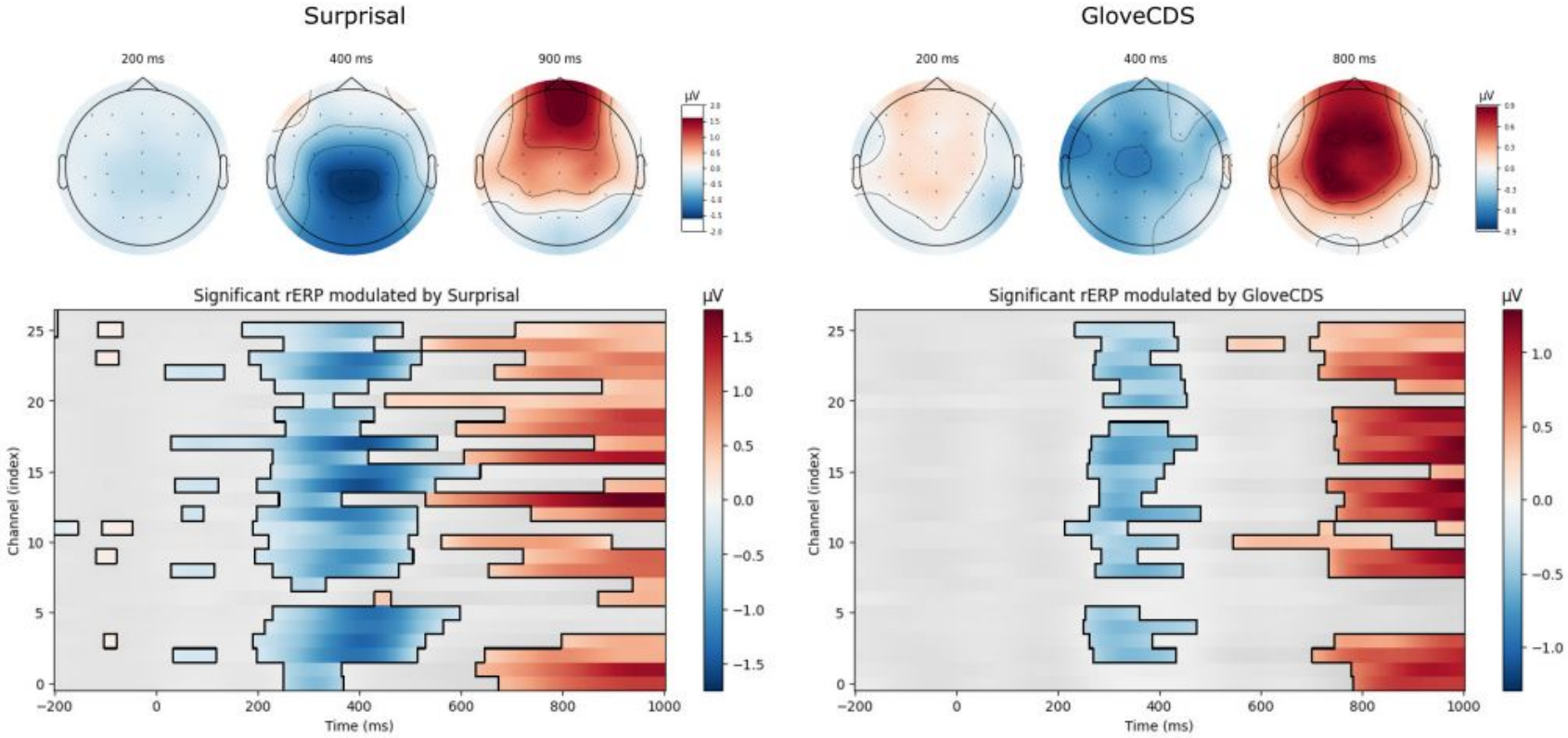
Zou, A., Phan, L., Chen, S., Campbell, J., Guo, P., Ren, R., ... & Hendrycks, D. (2023). Representation engineering: A top-down approach to ai transparency.

EEG signals vs LLM surprisal



Xu, H., Nakanishi, M., & Coulson, S. (2024). Revisiting Joke Comprehension with Surprisal and Contextual Similarity: Implication from N400 and P600 Components. In *Proceedings of the Annual Meeting of the Cognitive Science Society* (Vol. 46).

Interests?

Grading

See Course Organization document